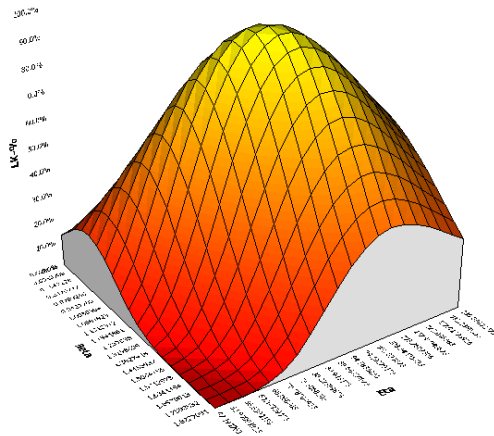


Likelihood Function Surface

Maximum Likelihood I

Lecture/Lab 09

Physics 4730, Fall 2026



Recap

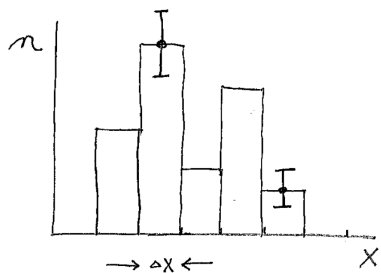
▶ Classical Inference (from data)

- ▶ Estimate parameters
- ▶ Quantify confidence
- ▶ Evaluate consistency of data and model

▶ Techniques

- ▶ Method of moments (mean, standard deviation, skewness, kurtosis ...)
- ▶ χ^2 -minimization/least squares “fitting” EPDF, for data with Gaussian uncertainties

Histogrammed Data



- Uncertainty in bin contents determined by *binomial statistics*:

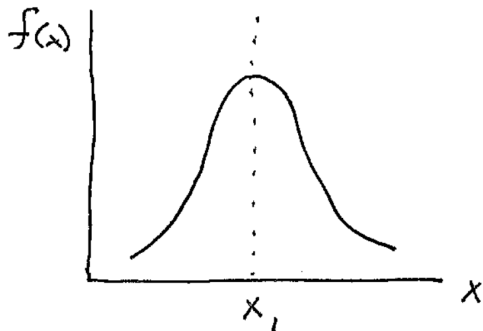
$$\begin{aligned}\sigma_{bin} &= \sqrt{Np(1-p)} \\ &= \sqrt{n} \text{ for small } p\end{aligned}$$

- Error in highly-populated bins well approximated by Gaussian with $\sigma = \sqrt{n}$.
- Error in low-population bins approximated by Poisson distribution **Not symmetric!**
- We need a method for fitting that **does not** assume Gaussian error!

Maximum Likelihood Estimation (MLE);

Qualitative Introduction:

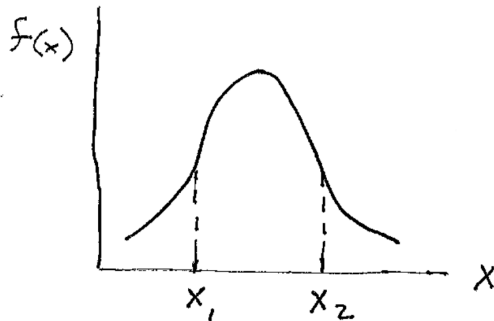
Suppose we have **one** data point drawn from a Gaussian RV. The best guess for the **mean** of that RV is x_1 .



→ The best guess for the *true* distribution is the one that **maximizes** $f(x_1)$.

Maximum Likelihood Estimation (MLE); Qualitative Introduction:

Suppose I've obtained **two**
points, x_1 and x_2 .



→ The best guess for the *true* distribution is
the one that **maximizes the product**
 $f(x_1) \cdot f(x_2)$.

Best Guess for N Measurements?

$$L = \prod_{i=1}^N f(x_i | M(\theta))$$

- ▶ L = “likelihood”
- ▶ $\prod$ = a product of terms
- ▶ f = a probability density function (PDF)
- ▶ x_i is the RV being measured
- ▶ $M(\theta)$ is a model dependent on parameter(s) θ

→ The best value(s) of θ will be that(those) which **Maximizes** L .

Example: Gaussian Distribution

Suppose I measure the length of a rod N times. Suppose these measurements have Gaussian, **homoscedastic** (all the same!) uncertainties. The likelihood is

$$L = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i - \mu)^2 / 2\sigma^2}$$

In this case, μ is the parameter we wish to determine. μ is ...

- ▶ ... the true length of the rod
- ▶ ... “ground truth”
- ▶ ... “truth with a capital T”

Example: Gaussian Distribution

Now, in practice L is a **very small number** and can be hard to compute. So instead we'll work with it's **natural logarithm**.

$$\begin{aligned}\ln(L) &= \sum_{i=1}^N \ln \left(\frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i - \mu)^2 / 2\sigma^2} \right) \\ &= N \ln \left(\frac{1}{\sqrt{2\pi}\sigma} \right) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} \\ &= [\text{constant}] - \frac{1}{2} \chi^2\end{aligned}$$

In other words, the maximum log-likelihood $\leftrightarrow$ minimum χ^2 !
(This is also true in the **heteroscedastic** case.)

Example: Gaussian Distribution

Now, to find **maximum likelihood**

$$\begin{aligned}\left(\frac{\partial}{\partial \mu} \ln(L)\right)_{\mu=\mu_0} &= 0 \\ -\frac{1}{2\sigma^2} \sum_{i=1}^N -2(x_i - \mu_0) &= 0 \\ \frac{1}{\sigma^2} \left(\sum_{i=1}^N x_i - \sum_{i=1}^N \mu_0 \right) &= \frac{1}{\sigma^2} \left(\sum_{i=1}^N x_i - N\mu_0 \right) = 0 \\ \mu_0 &= \frac{1}{N} \sum_{i=1}^N x_i\end{aligned}$$

→ μ_0 is the mean of all measurements!

Properties of MLE

- ▶ consistent convergence
- ▶ asymptotically normal (Gaussian)
- ▶ best possible estimate given data at hand

Example: Exponential Distribution

$$f(x|M(\beta)) = \frac{1}{\beta} e^{-x/\beta}$$

For a set $\{x_i\}$ of measurements, what is the most likely value of β ?

$$\begin{aligned} L &= \prod_{i=1}^N \frac{1}{\beta} e^{-x_i/\beta} \\ &= \frac{1}{\beta^N} \prod_{i=1}^N e^{-x_i/\beta} \end{aligned}$$

Now, $e^a e^b e^c \dots = e^{a+b+c+\dots}$, and

$$L = \frac{1}{\beta^N} e^{-(\sum x_i)/\beta}$$

Example: Exponential Distribution

$$\mathcal{L} = \ln(L) = -N \ln \beta - \frac{1}{\beta} N \bar{x}$$

$$\left. \frac{\partial \mathcal{L}}{\partial \beta} \right|_{\beta=\beta_0} = -\frac{N}{\beta_0} + \frac{N}{\beta_0^2} \bar{x} = 0$$

$$\beta_0 = \bar{x}$$

→ Recall that we used this result in Lab #7!

Please complete the calculations below in class. We will make use of the results in Lab 10.

1. The discrete Poisson probability density function is given by:

$$f(x|\lambda) = \frac{\lambda^x}{x!} e^{-\lambda}$$

where x is a nonnegative integer. Write down an expression for the *likelihood function* L for a set of measurements x_i of a Poisson random variable. Apply any simplifications you can think of.

Lab 9: Poisson Distribution

Submit worksheet to instructor

2. Now, find the *log-likelihood* function $\ln(L)$. Apply identities to break it into three terms: one dependent on x_i , one dependent on $x_i!$, and a constant.

3. Finally, by maximizing $\ln(L)$ with respect to λ , find an expression for the MLE for λ .